

A Solution Manual for
Introduction to Linear Algebra
by Serge Lang

Chèrmyn

Contents

I	Vectors	1
§1.	Definition of Points in Space	1
§2.	The Norm of a Vector	3
II	Matrices and Linear Equations	5
III	Vector Spaces	6
§1.		6
§2.		6
§3.	Convex Sets	6
§4.		7
§5.		7
§6.	The Rank of a Matrix	7
IV	Linear Mappings	9
§1.		9
§2.	Linear Mappings	9
§3.	The Kernel and Image of a Linear Map	10
§4.	The Rank and Linear Equations Again	12
§5.	The Matrix Associated with a Linear Map	14
V	Composition and Inverse Mappings	15
§1.	Composition of Linear Maps	15
§2.	Inverse	18
VI	Scalar Products and Orthogonality	20
§1.	Scalar Products	20
§2.	Orthogonal Bases	23
§3.	Bilinear Maps and Matrices	25
VII	Determinants	27
§1.	Determinants of Order 2	27
§2.	3×3 and $n \times n$ Determinants	27
§3.	The Rank of a Matrix and Subdeterminants	28
§4.	Cramer's Rule	28
§5.	Inverse of a Matrix	28
§6.	Determinants as Area and Volume	28

VIII Eigenvectors and Eigenvalues	30
§1. Eigenvectors and Eigenvalues	30
§2. The Characteristic Polynomial	32
§3. Eigenvalues and Eigenvectors of Symmetric Matrices	34
§4. Diagonalization of a Symmetric Linear Map	34

Acknowledgment

I would like to express my sincere appreciation to Professor Serge Lang for authoring such an excellent and illuminating introduction to linear algebra.

You might wonder: since Professor Serge Lang already provided answers to most exercises in *Introduction to Linear Algebra*, why write yet another solution manual?

My motivation is simple: to document my own journey of studying this classic text and capture the insights, struggles, and small victories along the way. While working through the book, I noticed that a few exercises lack official solutions. This manual aims to offer fellows using Lang's textbook some helpful reference and a bit of convenience.

A few clarifications about this manual:

1. Scope and Notation

I have not solved every exercise in the book — only those that particularly interested me. For these, I've written out complete, step-by-step solutions. I strive for logical rigor and clarity, always aligning with the conceptual framework and expository style of Lang's text, staying true to its axiomatic and structural approach.

This manual includes the partial solutions originally provided by Professor Lang, supplemented by my own (Chèrmyn's) complete solutions to selected exercises for which no answer was given. To distinguish sources, we adopt the following convention:

- (i) "Solution." (capitalized) denotes the answer or hint given by Professor Lang in the original text. Note that even when Lang's contribution is only a partial hint or intermediate step, it is still presented under "Solution".
- (ii) "solution." (lowercase) denotes a full solution authored by Chèrmyn.

2. Purpose Beyond Answers

More than just a collection of solutions, this manual reflects my appreciation for axiomatic mathematics. Through self-study, I aim to sharpen my logical reasoning and gradually build the foundation needed to engage with contemporary mathematical literature.

3. Community and Collaboration

I warmly welcome corrections, alternative approaches, or deeper insights from readers. Let us learn and grow together on this beautiful path of mathematical discovery.

This manual is strictly non-commercial and intended solely for personal study and academic exchange. All exercise statements are reproduced from Serge Lang's *Introduction to Linear Algebra*, and the copyright of which remain the property of the author and publisher.

During my reading, I noticed a few minor typographical or notational slips — nothing that affects the mathematical correctness, but occasionally enough to cause momentary confusion. My aim in noting these is simply to help future readers follow Lang's elegant exposition more smoothly, without stumbling over avoidable notational quirks. Excluding insignificant cases such as the zero vector being written as “0” instead of “O”, which do not affect understanding, corrections are marked in red and listed below:

1. Chapter II, §3, p. 64: “We shall find three ways of interpreting such equations, illustrating three different ways of thinking about matrices and vectors.”
2. Chapter II, §4, p. 74: “One could make further row operations to insert further zeros, for instance subtract the second row from the first, and then add $\frac{4}{5}$ times the third row to the second. This yields:

$$\begin{pmatrix} 1 & 0 & -\frac{1}{5} & 0 & \frac{2}{5} \\ 0 & 1 & 0 & -\frac{17}{13} & -\frac{3}{13} \\ 0 & 0 & 1 & -\frac{5}{13} & \frac{19}{13} \end{pmatrix}.”$$

3. Chapter IV, §5, p. 155: “The change of coordinates from one basis to another is given in terms of the transpose of the matrix expressing each w_i as a linear combination of $v_1, \dots, v_n$.”
4. Chapter VII, §2, p. 205: “In the matrix A we replace the j -th column and the k -th column by $A^j + A^k$.”
5. Chapter VII, §3, p. 211: “ $= \sum_{k \neq j} c_k D(A^1, \dots, A^k, \dots, A^n)$ ”.
6. Chapter VIII, Appendix, p. 262: “ $= |a|^2 + 2|\beta||\alpha| + |\beta|^2$ ”.
7. Answers to Exercises, V, §1, p. 279, Exercise 5: “Then $Qv = QPw = 0$ because $QP = (I - P)P = P - P^2 = P - P = O$.”
8. Answers to Exercises, VIII, §4, p. 288, Exercise 3: “Then $\langle Av, v \rangle = \langle \lambda v, v \rangle = \lambda \langle v, v \rangle$.”
9. Answers to Exercises, VIII, §4, p. 288, Exercise 6: “ $= \langle (v + w), (v + w) \rangle - \langle (v - w), (v - w) \rangle$ ”.

Chapter I

Vectors

§1. Definition of Points in Space

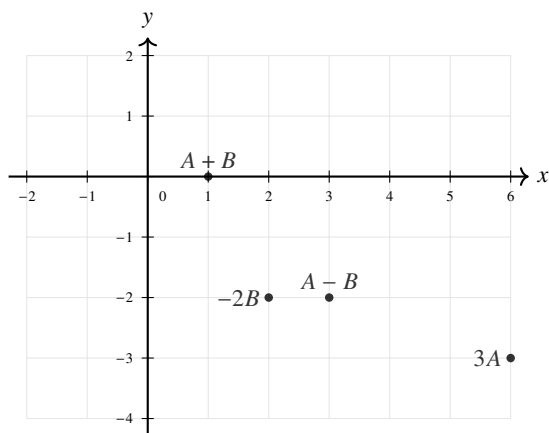
Exercises I, §1.

Find $A + B$, $A - B$, $3A$, $-2B$ in each of the following cases. Draw the points of Exercises 1 and 2 on a sheet of graph paper.

1. $A = (2, -1)$, $B = (-1, 1)$

Solution.

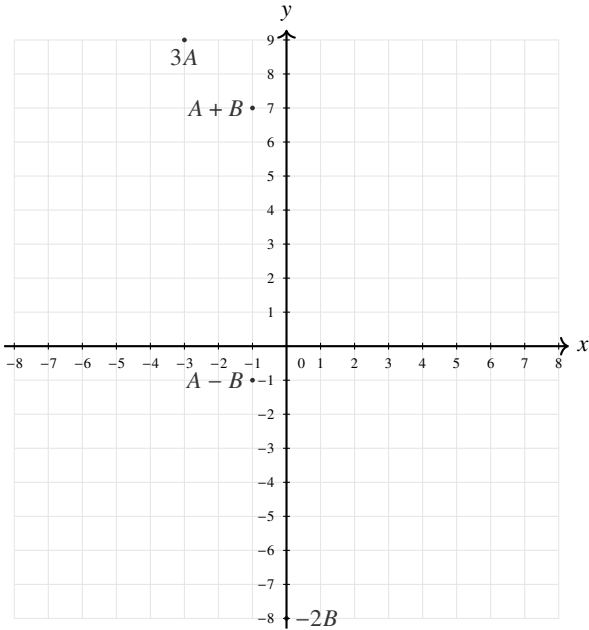
$A + B$	$A - B$	$3A$	$-2B$
$(1, 0)$	$(3, -2)$	$(6, -3)$	$(2, -2)$



2. $A(-1, 3), B(0, 4)$

Solution.

$A + B$	$A - B$	$3A$	$-2B$
$(-1, 7)$	$(-1, -1)$	$(-3, 9)$	$(0, -8)$



3. $A(2, -1, 5), B(-1, 1, 1)$

Solution.

$A + B$	$A - B$	$3A$	$-2B$
$(1, 0, 6)$	$(3, -2, 4)$	$(6, -3, 15)$	$(2, -2, -2)$

4. $A(-1, -2, 3), B(-1, 3, -4)$

Solution.

$A + B$	$A - B$	$3A$	$-2B$
$(-2, 1, -1)$	$(0, -5, 7)$	$(-3, -6, 9)$	$(2, -6, 8)$

5. $A(\pi, 3, -1), B(2\pi, -3, 7)$

Solution.

$A + B$	$A - B$	$3A$	$-2B$
$(3\pi, 0, 6)$	$(-\pi, 6, -8)$	$(3\pi, 9, -3)$	$(-4\pi, 6, -14)$

6. $A(15, -2, 4), B(\pi, 3, -1)$

Solution.

$A + B$	$A - B$	$3A$	$-2B$
$(15 + \pi, 1, 3)$	$(15 - \pi, -5, 5)$	$(45, -6, 12)$	$(-2\pi, -6, 2)$

§2. The Norm of a Vector

Exercises I, §2.

- 1.
- 2.
- 3.
- 4.
- 5.
- 6.
- 7.

8. For any vectors A, B , prove the following relations:

(a) $\|A + B\|^2 + \|A - B\|^2 = 2\|A\|^2 + 2\|B\|^2$.

solution.

Proof. To prove this, we use the definitions, namely

$$\begin{aligned}\|A + B\|^2 + \|A - B\|^2 &= (A + B) \cdot (A + B) + (A - B) \cdot (A - B) \\ &= A^2 + 2A \cdot B + B^2 + A^2 - 2A \cdot B + B^2 \\ &= 2A^2 + 2B^2 \\ &= 2\|A\|^2 + 2\|B\|^2,\end{aligned}$$

because $A^2 = \|A\|^2$ and $B^2 = \|B\|^2$ by definition. Then we conclude the proof. $\square$

(b) $\|A + B\|^2 = \|A\|^2 + \|B\|^2 + 2A \cdot B$.

(c) $\|A + B\|^2 - \|A - B\|^2 = 4A \cdot B$.

Interpret (a) as a “parallelogram law”.

solution.

Consider vectors A and B as adjacent sides of a parallelogram (with a common initial point). By definitions, $A + B$ and $A - B$ correspond to two diagonals of the parallelogram. Thus (a) means that the sum of the squares of the lengths of the two diagonals of a parallelogram equals twice the sum of the squares of the lengths of its adjacent sides, which is exactly the “parallelogram law”.

Chapter II

Matrices and Linear Equations

Chapter III

Vector Spaces

§1.

§2.

§3. Convex Sets

Exercises III, §3.

1.

2. Let A be a non-zero vector in $\mathbf{R}^n$ and let c be a fixed number. Show that the set of all elements X in $\mathbf{R}^n$ such that $A \cdot X \geq c$ is convex.

solution.

Proof. Let the set of all elements X in $\mathbf{R}^n$ such that $A \cdot X \geq c$ be denoted by S .

We wish to prove that S is convex. Let X_1 and X_2 be points in S , then a point between them can be written as

$$(1-t)X_1 + tX_2.$$

Hence

$$\begin{aligned} A \cdot [(1-t)X_1 + tX_2] &= (1-t)(A \cdot X_1) + t(A \cdot X_2) \\ &\geq (1-t)c + tc \\ &= c \end{aligned}$$

This proves that $(1-t)X_1 + tX_2$ is in the set S , which is therefore convex. $\square$

§4.

§5.

§6. The Rank of a Matrix

Exercises III, §6.

- 1.
- 2.
3. Let A be an $m \times n$ matrix and let B be an $n \times r$ matrix, so we can form the product AB .
 - (a) Show that the columns of AB are linear combinations of the columns of A . Thus prove that

$$\text{rank } AB \leq \text{rank } A.$$

- (b) Prove that $\text{rank } AB \leq \text{rank } B$. [*Hint*: Use the fact that

$$\text{rank } AB = \text{rank } {}^t(AB)$$

and

$$\text{rank } B = \text{rank } {}^tB.]$$

solution.

- (a) We first show that the columns of AB are linear combinations of the columns of A .

Proof. Let $A^1, A^2, \dots, A^n$ be the columns of A , and matrix B is of the form

$$\begin{pmatrix} b_{11} & b_{12} & \cdots & b_{1r} \\ b_{21} & b_{22} & \cdots & b_{2r} \\ \vdots & \vdots & \ddots & \vdots \\ b_{n1} & b_{n2} & \cdots & b_{nr} \end{pmatrix}$$

Let $(AB)^1, (AB)^2, \dots, (AB)^r$ be the columns of AB , by the definition of matrix multiplication, we have

$$(AB)^j = b_{1j}A^1 + b_{2j}A^2 + \cdots + b_{nj}A^n \quad \text{for } j = 1, 2, \dots, r,$$

which proves the columns of AB are linear combinations of the columns of A as desired. $\square$

Then we continue our proof.

Proof. From the proof above, we observe that any column of AB is a linear combination of $A^1, A^2, \dots, A^n$. So do any linear combination of the columns of AB . Consequently the column space of AB is contained in the column space of A . Hence by Theorem 5.8, we have

$$\text{rank } AB \leq \text{rank } A$$

as desired. $\square$

- (b) *Proof.* The transpose of the product is equal to the product of the transpose in the reverse order, that is to say

$$\text{rank}^t(AB) = \text{rank}^t B^t A.$$

Using the result of (a), we have

$$\text{rank}^t B^t A \leq \text{rank}^t B.$$

From the hint, we see

$$\text{rank } AB = \text{rank}^t(AB) \quad \text{and} \quad \text{rank } B = \text{rank}^t B.$$

Combining three expressions above, we conclude

$$\text{rank } AB \leq \text{rank } B$$

as desired. $\square$

Chapter IV

Linear Mappings

§1.

§2. Linear Mappings

Exercises IV, §2.

- 1.
- 2.
- 3.
- 4.
- 5.
- 6.
- 7.
- 8.
- 9.
- 10.
- 11.
- 12.
- 13.
14. Show that the image of a convex set under a linear map is convex.
solution.

Proof. Let $L : V \rightarrow W$ be a linear map, and let S be a convex set in V . We denote the image of S under L by $L(S)$.

Let $L(v)$ and $L(w)$ be two points in $L(S)$ such that $v, w \in S$. Let $0 \leq t \leq 1$. Then

$$tL(v) + (1-t)L(w) = L[tv + (1-t)w].$$

Then $tv + (1-t)w$ lies in convex set S , whence $L[tv + (1-t)w]$ is a point in $L(S)$. From the definition of convex set, we show that the image of a convex set under a linear map is convex as desired. $\square$

§3. The Kernel and Image of a Linear Map

Exercises IV, §3.

- 1.
- 2.
- 3.
- 4.
5. Let $L : V \rightarrow W$ be a linear map. Let w be an element of W . Let v_0 be an element of V such that $L(v_0) = w$. Show that any solution of the equation $L(X) = w$ is of type $v_0 + u$, where u is an element of the kernel of L .

Solution.

Proof. Since $L(v_0 + u) = L(v_0)$ if u is in $\text{Ker } L$, every element of the form $v_0 + u$ is a solution. Conversely, let v be a solution of $L(v) = w$. Then

$$L(v - v_0) = L(v) - L(v_0) = w - w = 0,$$

so $v - v_0 = u$ is in the kernel, and $v = v_0 + u$. $\square$

6. Let V be the vector space of functions which have derivatives of all orders, and let $D : V \rightarrow V$ be the derivative. What is the kernel of D ?

solution.

Constant functions.

- 7.
- 8.
- 9.
- 10.

- 11.
- 12.
- 13.
- 14.
- 15.
16. (To be done after you have done Exercise 15.) Let U, W be subspaces of a vector space V . Show by the indicated method that

$$\dim U + \dim W = \dim(U + W) + \dim(U \cap W).$$

- (a) Show that the map

$$L : U \times W \rightarrow V$$

given by

$$L(u, w) = u - w$$

is a linear map.

- (b) Show that image of L is $U + W$.
- (c) Show that the kernel of L is the subspaces of $U \times W$ consisting of all elements (u, u) where u is in $U \cap W$. What is a basis for this subspace?
- (d) Apply the dimension formula in the text to conclude the proof.

Solution.

- (a) *Proof.* For any elements $(u_1, w_1), (u_2, w_2)$ of $U \times W$, we have

$$\begin{aligned} L((u_1 + w_1) + (u_2 + w_2)) &= L(u_1 + u_2, w_1 + w_2) \\ &= (u_1 + u_2) - (w_1 + w_2) \\ &= (u_1 - w_1) + (u_2 - w_2) \\ &= L(u_1, w_1) + L(u_2, w_2). \end{aligned}$$

Furthermore, if c is a number, then

$$\begin{aligned} L(c(u_1, w_1)) &= L(cu_1, cw_1) \\ &= cu_1 - cw_1 \\ &= c(u_1 - w_1) \\ &= cL(u_1, w_1) \end{aligned}$$

Hence L is a linear map. □

- (b) *Proof.* The image is clearly contained in $U + W$. Given an arbitrary element $u + w$ with u in U and w in W , we can write it in the form $u + w = u - (-w)$, which shows that it is in the image of L . □
- (c) *Proof.* The kernel of L consists of those elements (u, w) such that $u - w = 0$, so $u = w$. In other words, it consists of the pairs (u, u) , and u must lie both in U and W , so in the intersection. If $\{u_1, \dots, u_r\}$ is a basis for $U \cap W$, then $\{(u_1, u_1), \dots, (u_r, u_r)\}$ is a basis for the kernel of L . □

- (d) From (b), we see that the image of L is $U + W$, that is to say

$$\dim \operatorname{Im} L = \dim(U + W).$$

From (c), we see the dimension of the kernel of L is the same as the dimension of $U \cap W$, that is to say

$$\dim \operatorname{Ker} L = \dim(U \cap W).$$

Applying the dimension formula in the text

$$\dim L = \dim \operatorname{Ker} L + \dim \operatorname{Im} L,$$

then we have

$$\dim(U \times W) = \dim(U \cap W) + \dim(U + W).$$

Substituting the dimension of $U \times W$ which we have already shown before by

$$\dim(U \times W) = \dim U + \dim W.$$

Hence we conclude the proof

$$\dim U + \dim W = \dim(U + W) + \dim(U \cap W).$$

§4. The Rank and Linear Equations Again

Exercises IV, §4.

1. Let A be a non-zero vector in $\mathbf{R}^n$. What is the dimension of the space of solution of the solutions of the equation $A \cdot X = 0$?

Solution.

$n-1$

2. What is the dimension of the subspace of $\mathbf{R}^6$ perpendicular to the two vectors $(1, 1, -2, 3, 4, 5)$ and $(0, 0, 1, 1, 0, 7)$?

Solution.

4

3. Let A be a non-zero vector in n -space. Let P be a point in n -space. What is the dimension of the set of solutions of the equation

$$X \cdot A = P \cdot A?$$

Solution.

$n-1$

4.

5.

6. Let $L : V \rightarrow W$ be a linear map. Using a theorem from the text, prove that

$$\dim \operatorname{Im} L \leq \dim V.$$

Solution.*Proof.* One theorem states that

$$\dim V = \dim \operatorname{Im} L + \dim \operatorname{Ker} L.$$

Since $\dim \operatorname{Ker} L \geq 0$, the desired inequality follows. $\square$ 7. Let A, B be two matrices which can be multiplied, i.e. such that AB exists. Prove that

$$\operatorname{rank} AB \leq \operatorname{rank} A \quad \text{and} \quad \operatorname{rank} AB \leq \operatorname{rank} B.$$

Solution.

One proof (there are others): $\operatorname{rank} A = \dim \operatorname{Im} L_A$. But $L_{AB} = L_A \circ L_B$. Hence the image of L_{AB} is contained in the image of L_A . Hence $\operatorname{rank} AB \leq \operatorname{rank} A$.

For the other inequality, note that the rank of a matrix is equal to the rank of its transpose, because column rank equals row rank. Hence

$$\operatorname{rank} AB = \operatorname{rank} {}^t B {}^t A.$$

Now apply the first inequality to get $\operatorname{rank} {}^t B {}^t A \leq \operatorname{rank} {}^t B = \operatorname{rank} B$.8. Let U be a subspace of $\mathbf{R}^n$. Prove that $U^\perp$ is also a subspace.**solution.**

Proof. Observe first that O is the vector which is perpendicular to all elements of U , and hence O is in the orthogonal complement of U , i.e. $U^\perp$.

Next, suppose that u_1, u_2 are in the $U^\perp$. Then for arbitrary element u of U , we have

$$(u_1 + u_2) \cdot u = u_1 \cdot u + u_2 \cdot u = 0,$$

thereby proving that $u_1 + u_2$ is in the orthogonal complement of U .If c is a number, then

$$cu_1 \cdot u = c(u_1 \cdot u) = 0.$$

Hence cu_1 is in the orthogonal complement of U . This proves that $U^\perp$ is also a subspace. $\square$

§5. The Matrix Associated with a Linear Map

Exercises IV, §5.

- 1.
2. Let c be a number, and let $L : R^n \rightarrow R^n$ be the linear map such that $L(X) = cX$. What is the matrix associated with this linear map?

Solution.

cI , where I is the unit $n \times n$ matrix.

- 3.
- 4.
5. In the text, we gave a description of a matrix associated with a linear map of a vector space into itself, with respect to a basis. More generally, let V, W be two vector spaces. Let $\{v_1, \dots, v_n\}$ be a basis of V , and $\{w_1, \dots, w_m\}$ be a basis of W . Let $L : V \rightarrow W$ be a linear map. Describe how you would associate a matrix with this linear map, giving the effect on coordinate vectors.

Solution.

Let $Lv_i = \sum c_{ij}w_j$. Let $C = (c_{ij})$. The associated matrix is tC , and the effect of L on a coordinate vector X is tCX .

- 6.
7. Let V be the vector space generated by the two functions $f_1(t) = \cos t$ and $f_2(t) = \sin t$. Let D be the derivative. What is the matrix of D with respect to the basis $\{f_1, f_2\}$?

Solution.

$$\begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}$$

Chapter V

Composition and Inverse Mappings

§1. Composition of Linear Maps

Exercises V, §1.

1. Let A, B be two $m \times n$ matrices. Assume that

$$AX = BX$$

for all n -tuples X . Show that $A = B$.

Solution.

Proof. Let $C = A - B$. Then $CX = O$ for all X . Take $X = E^j$ to be the j -th unit vector for $j = 1, \dots, n$. Then $CE^j = C^j$ is the j -th column of C . By assumption, $CE^j = O$ for all j so $C = O$. $\square$

2.

3.

4.

5. Let $P : V \rightarrow V$ be a linear map such that $P^2 = P$. Define $Q = I - P$.

(a) Show that $Q^2 = Q$.

(b) Show that $\text{Im } P = \text{Ker } Q$ and $\text{Ker } P = \text{Im } Q$.

A linear map P such that $P^2 = P$ is called a **projection**. It generalizes the notion of projection in the usual sense.

Solution.

(a) *Proof.*

$$Q^2 = (I - P)^2 = I^2 - 2IP + P^2 = I - 2P + P = I - P.$$

□

(b) *Proof.* Let $v \in \text{Im } P$ so $v = Pw$ for some w . Then $Qv = QPw = 0$ because $QP = (I - P)P = P - P^2 = P - P = 0$. Hence $\text{Im } P \subset \text{Ker } Q$. Conversely, let $v \in \text{Ker } Q$ so $Qv = 0$. Then $(I - P)v = 0$ so $v - Pv = 0$, and $v = Pv$, so $v \in \text{Im } P$, whence $\text{Ker } Q \subset \text{Im } P$. Therefore we conclude that $\text{Im } P = \text{Ker } Q$.

Similarly, we can show that $\text{Ker } P = \text{Im } Q$. Let $v \in \text{Ker } P$ so $Pv = 0$. Then $(I - Q)v = 0$ so $v - Qv = 0$, and $v = Qv$, so $v \in \text{Im } Q$, whence $\text{Ker } P \subset \text{Im } Q$. Conversely, let $v \in \text{Im } Q$ so $v = Qw$ for some w . Then $Pv = PQw = 0$ because $PQ = P(I - P) = P - P^2 = P - P = 0$. Hence $\text{Im } Q \subset \text{Ker } P$. Therefore we conclude that $\text{Ker } P = \text{Im } Q$. □

6. Let $P : V \rightarrow V$ be a **projection**, that is the linear map such that $P^2 = P$.

(a) Show that $V = \text{Ker } P + \text{Im } P$.

(b) Show that the intersection of $\text{Im } P$ and $\text{Ker } P$ is $\{0\}$. In other words, if $v \in \text{Im } P$ and $v \in \text{Ker } P$, then $v = 0$.

Solution.

(a) *Proof.* Let $v \in V$. Then $v = v - Pv + Pv$, and $v - Pv \in \text{Ker } P$ because

$$P(v - Pv) = Pv - P^2v = Pv - Pv = 0.$$

Furthermore $Pv \in \text{Im } P$, thus proving (a). □

(b) *Proof.* Let $v \in \text{Im } P \cap \text{Ker } P$. Since $v \in \text{Im } P$ there exists $w \in V$ such that $v = Pw$. Since $v \in \text{Ker } P$, we get

$$0 = Pv = P^2w = Pw = v,$$

so $v = 0$, whence (b) is also proved. □

Let V be a vector space, and let U, W be subspaces. One says that V is a **direct sum** of U and W if the following conditions are satisfied:

$$V = U + W \quad \text{and} \quad U \cap W = \{0\}.$$

In Exercise 6, we have proved that V is the direct sum of $\text{Ker } P$ and $\text{Im } P$.

7.

8.

9. Let A, B be two matrices so that the product AB is defined. Prove that

$$\text{rank } AB \leq \text{rank } A \quad \text{and} \quad \text{rank } AB \leq \text{rank } B.$$

[Hint: Consider the linear maps L_{AB}, L_A , and L_B .]

solution.

Proof. Let L_{AB} , L_A and L_B be the linear maps such that

$$L_{AB}(X) = ABX, \quad L_A(X) = AX \quad \text{and} \quad L_B(X) = BX.$$

By composition of linear maps corresponding to multiplication of matrices, we have

$$L_{AB} = L_A \circ L_B.$$

The rank of the matrix is the dimension of the image of its linear map, that is to say

$$\text{rank } AB = \dim \text{Im } L_{AB}, \quad \text{rank } L_A = \dim \text{Im } A \quad \text{and} \quad \text{rank } B = \dim \text{Im } B.$$

We first show that $\text{rank } AB \leq \text{rank } A$. It is clear that $\text{Im } L_{AB} \subseteq \text{Im } L_A$. Then

$$\dim \text{Im } L_{AB} \leq \dim \text{Im } L_A,$$

because the dimension of the subspace of a space is less than or equal to the dimension of that space. Then we can immediately conclude that

$$\text{rank } AB \leq \text{rank } A.$$

Next, we show that $\text{rank } AB \leq \text{rank } B$. We have already shown before that the dimension of the image of a linear map is less than or equal to the dimension of its domain, say

$$\dim \text{Im } L_{AB} = \dim \text{Im}(L_A \circ L_B) \leq \dim \text{Im } L_B,$$

because we can regard L_A as being restricted to the subspace $\text{Im } L_B$ in the composition $L_{AB} = L_A \circ L_B$. Then we sort the expressions and then conclude that

$$\text{rank } AB \leq \text{rank } B.$$

Hence we complete the proof. $\square$

In fact, define the **rank of a linear map** L to be $\dim \text{Im } L$. If $L : V \rightarrow W$ and $F : W \rightarrow U$ are linear maps of finite dimensional spaces, show that

$$\text{rank } F \circ L \leq \text{rank } F \quad \text{and} \quad \text{rank } F \circ L \leq \text{rank } L.$$

Solution.

Proof. The dimension of a subspace is $\leq$ the dimension of the space. Then

$$\text{Im } F \circ L \leq \text{Im } F \quad \text{so} \quad \dim \text{Im } F \circ L \leq \dim \text{Im } F$$

so $\text{rank } F \circ L \leq \text{rank } F$. This proves one of the formulas. For the other, view F as a linear map defined on $\text{Im } L$. Then

$$\text{rank } F \circ L = \dim \text{Im } F \circ L \leq \dim \text{Im } L = \text{rank } L.$$

$\square$

§2. Inverse

Exercises V, §2.

1. Let R_θ be rotation counterclockwise by an angle θ . How would you express the inverse R_θ^{-1} as R_φ for some φ in a simple way? If

$$A(\theta) = \begin{pmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{pmatrix}$$

is the matrix associated with R_θ , what is the matrix associated with R_θ^{-1} ?

Solution.

$R_\theta^{-1} = R_{-\theta}$ because $R_\theta \circ R_{-\theta} = R_{\theta-\theta} = R_0 = I$. The matrix associated with R_θ^{-1} is

$$\begin{pmatrix} \cos \theta & \sin \theta \\ -\sin \theta & \cos \theta \end{pmatrix}$$

because $\cos(-\theta) = \cos \theta$.

2. (a) Finish the proof of Theorem 2.1.
(b) Give the proof of Theorem 2.4 (ii).

solution.

- (a) *Proof.* Let $v \in V$. Then we must show that

$$G(cv) = cG(v).$$

Let $u = G(v)$. By definition, this means that

$$F(u) = F(G(v)) = (F \circ G)(v) = I_V v = v.$$

Since F is linear, we find that

$$F(cu) = cF(u) = cv.$$

By definition of the inverse map, this means that

$$G(cv) = G(F(cu)) = (G \circ F)(cu) = I_U(cu) = cv,$$

thus proving what we wanted. □

- (b) *Proof.* Suppose first that F is surjective. Then by the definition of surjection, we have

$$\text{Im } F = W \quad \text{so} \quad \dim \text{Im } F = \dim W.$$

But

$$\dim V = \dim \text{Ker } F + \dim \text{Im } F \quad \text{and} \quad \dim V = \dim W,$$

so $\dim \text{Ker } F = 0$, which implicates that $\text{Ker } F = \{O\}$. Hence F is injective by Theorem 2.2. This proves (ii), by using Theorem 2.3. □

- 3.
- 4.
- 5.
- 6.
- 7.
- 8.
- 9.
- 10.

11. Let V be a two-dimensional vector space, and let $L : V \rightarrow V$ be a linear map such that $L^2 = O$, but $L \neq O$. Let v be an element of V such that $L(v) \neq O$. Let $w = L(v)$. Prove that $\{v, w\}$ is a basis of V .

Solution.

Proof. It suffices to prove that v and w are linearly independent. Suppose $xv + yw = O$. Apply L . Then

$$\begin{aligned}
 L(O) &= L(xv + yw) \\
 &= xL(v) + yL(w) \\
 &= xL(v) + yL(L(v)) \\
 &= xL(v) + yO \\
 &= O
 \end{aligned}$$

because $L^2 = O$. Hence $xL(v) = O$. Since $L(v) \neq O$, it follows that $x = 0$. Then $y = 0$ because $w \neq O$. □

- 12.
- 13.

Chapter VI

Scalar Products and Orthogonality

§1. Scalar Products

bx

1. Let A be a symmetric $n \times n$ matrix. Given two column vectors $X, Y \in \mathbf{R}^n$, define

$$\langle X, Y \rangle = {}^tXAY.$$

- (a) Show that this symbol satisfies the first three properties of a scalar product.
- (b) Given an example of a 1×1 matrix and a non-zero 2×2 matrix such that the fourth property is not satisfied. If this fourth property is satisfied, that is ${}^tXAX > 0$ for all $X \neq O$, then the matrix A is called **positive definite**.
- (c) Given an example of 2×2 matrix which is symmetric and positive definite.
- (d) Let $a > 0$, and let

$$A = \begin{pmatrix} a & b \\ b & d \end{pmatrix}.$$

Prove that A is positive definite if and only if $ad - b^2 > 0$. [Hint: Let $X = {}^t(x, y)$ and complete the square in the expression tXAX .]

- (e) If $a < 0$ show that A is not positive definite.

Solution.

- (a) *Proof.* We first observe the result of the scalar product we defined is a 1×1 matrix, which means that it is simply a scalar.

We begin by showing the first property. We know

$$\langle X, Y \rangle = {}^tXAY \quad \text{and} \quad \langle Y, X \rangle = {}^tYAX.$$

And we observe that

$${}^t({}^tXAY) = {}^tY{}^tAX = {}^tYAX$$

because A is symmetric. Then we have

$$\langle X, Y \rangle = {}^t\langle X, Y \rangle = \langle Y, X \rangle$$

because $\langle X, Y \rangle$ and $\langle Y, X \rangle$ are both scalars, which proving the first property.

We continue to show the second property. Let $Z \in \mathbf{R}^n$ is a column vector. Then by the distributivity of matrices, we have

$$\langle X, Y + Z \rangle = {}^tXA(Y + Z) = {}^tXAY + {}^tXAZ = \langle X, Y \rangle + \langle X, Z \rangle,$$

which proves the second property.

We finally come to the third property. If x is a number, then

$$\langle xX, Y \rangle = {}^t(xX)AY = x{}^t(X)AY = x\langle X, Y \rangle.$$

On the other hand, we have

$$\langle X, xY \rangle = {}^tXA(xY) = x{}^tXAY = x\langle X, Y \rangle.$$

Thus we conclude the whole proof. □

(b) $(-1), \begin{pmatrix} -1 & 0 \\ 0 & -1 \end{pmatrix}$

(c) $\begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$

(d) *Proof.* Let $X = {}^t(x, y)$. Then

$$\langle X, X \rangle = ax^2 + 2bxy + dy^2 = a \left(x + \frac{b}{a}y \right)^2 + \frac{(ad - b^2)}{a}y^2.$$

If $ad - b^2 > 0$, then $\langle X, X \rangle$ is a sum of squares with positive coefficients, and one of the two terms is not 0, so $\langle X, X \rangle > 0$. If $ad - b^2 \leq 0$, then give y any non-zero value, and let $x = -by/a$. Then $\langle X, X \rangle \leq 0$. □

(e) *Proof.* Let $X = {}^t(1, 0) \neq O$ (of course). Then

$$\langle X, X \rangle = a < 0,$$

which contradicts the definition of the positive definite matrix. Hence $a < 0$ implies A is not positive definite. □

2.

3.

4. Let A be an $n \times n$ matrix. Define the **trace** of A to be the sum of the diagonal elements. Thus if $A = (a_{ij})$, then

$$\text{tr}(A) = \sum_{i=1}^n a_{ii}.$$

For instance, if

$$A = \begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix},$$

then $\text{tr}(A) = 1 + 4 = 5$. If

$$A = \begin{pmatrix} 1 & -1 & 5 \\ 2 & 1 & 3 \\ 1 & -4 & 7 \end{pmatrix},$$

then $\text{tr}(A) = 9$.

5.

6. If A is a symmetric square matrix, show that $\text{tr}(AA) \geq 0$, and $= 0$ if and only if $A = O$.

Solution.

Proof. Let $AA = (c_{ij})$. Then

$$c_{ii} = \sum_{k=1}^n a_{ik}a_{ki}.$$

Hence

$$\text{tr}(AA) = \sum_{i=1}^n \sum_{k=1}^n a_{ik}a_{ki}.$$

and $a_{ik}a_{ki} = a_{ik}^2$ because A is symmetric, so the trace is a sum of squares, hence ≥ 0 . $\square$

7.

8. (a) Prove in general that if A, B are square $n \times n$ matrices, then

$$\text{tr}(AB) = \text{tr}(BA).$$

- (b) If C is an $n \times n$ matrix which has an inverse, then $\text{tr}(C^{-1}AC) = \text{tr}(A)$.

Solution.

- (a) *Proof.*

$$\text{tr}(AB) = \sum_i \sum_j a_{ij}b_{ji} = \sum_j \sum_i b_{ji}a_{ij}$$

But any pair of letters can be used to denote indices in this sum, which can be rewritten more neutrally.

$$\sum_{r=1}^n \sum_{s=1}^n b_{rs}a_{sr}.$$

This is precisely the trace $\text{tr}(BA)$. $\square$

- (b) $\text{tr}(C^{-1}AC) = \text{tr}(ACC^{-1})$ by part (a), so $= \text{tr}(A)$.

§2. Orthogonal Bases

Exercises VI, §2.

- 1.
- 2.
- 3.
- 4.
- 5.
- 6.
7. Let V be a finite dimensional vector space with a positive definite scalar product. Let W be a subspace. Show that

$$V = W + W^\perp \quad \text{and} \quad W \cap W^\perp = \{O\}.$$

In the terminology of the preceding chapter, this means that V is the direct sum of W and its orthogonal complement. [Use Theorem 2.3.]

solution.

Proof. To the first identity, let $\{w_1, \dots, w_r\}$ be an orthogonal basis for W . Then we can extend it to an orthogonal basis $\{w_1, \dots, w_r, u_1, \dots, u_s\}$ for V by Theorem 2.1. Let U be the subspace generated by $u_1, \dots, u_s$. Then $U = W^\perp$ by Theorem 2.3.

Let $v \in V$. There exist numbers a_i ($i = 1, \dots, r$) and b_i ($i = 1, \dots, s$) such that

$$v = \sum_{i=1}^r a_i w_i + \sum_{j=1}^s b_j u_j.$$

We can regard the first summation as an element w of W and the second summation as an element u of U , which also implies the second sum is an element of $W^\perp$. Then

$$v = w + u,$$

where w and u are elements of W and $W^\perp$, respectively. Hence we conclude that

$$V = W + W^\perp.$$

Next, we come to the second identity. Let v be an element of $W \cap W^\perp$, which implicates v is simultaneously an element of W and an element of $W^\perp$. Then we have

$$\langle v, v \rangle = 0.$$

Since V is a finite dimensional vector space with a positive definite scalar product, we have $v = O$ if and only if $\langle v, v \rangle = 0$. Hence we conclude that $W \cap W^\perp = \{O\}$, which completes the proof. $\square$

8. In Exercise 7, show that $(W^\perp)^\perp = W$. Why is this immediate from Theorem 2.3?

solution.

Proof. We shall prove that $W \subset (W^\perp)^\perp$ and $(W^\perp)^\perp \subset W$, so $W = (W^\perp)^\perp$.

First let $w \in W$. Since w is perpendicular to all elements of $W^\perp$, we have

$$\langle w, v \rangle = 0,$$

where v is an arbitrary element of $W^\perp$. By the definition of orthogonal complement, we know w is an element of $(W^\perp)^\perp$. Thus we conclude

$$W \subset (W^\perp)^\perp.$$

Conversely, by Theorem 2.3, we have

$$\dim W + \dim W^\perp = \dim V \quad \text{and} \quad \dim W^\perp + \dim (W^\perp)^\perp = \dim V.$$

Subtracting the two equations above, then

$$\dim W = \dim (W^\perp)^\perp.$$

By Theorem 5.6 of Chapter III and the part above, we conclude

$$W = (W^\perp)^\perp.$$

□

It is immediate from Theorem 2.3, because we can also use the theorem to $W^\perp$ to get the result we desired.

9. (a) Let V be the space of symmetric $n \times n$ matrices. For $A, B \in V$ define

$$\langle A, B \rangle = \text{tr}(AB),$$

where tr is the trace (sum of diagonal elements). Show that this satisfies all the properties of a positive definite scalar product. (You might already have done this as an exercise in a previous section.)

- (b) Let W be the subspace of matrices A such that $\text{tr}(A) = 0$. What is the dimension of the orthogonal complement of W , relative to the scalar product in part (a)? Given an explicit basis for this orthogonal complement.

Solution.

- (a) *Proof.* We have proved the first property in the Exercise 8 (a) of Chapter VI before, so we start from the second property. Let $C \in V$. Then by the distributivity of matrices, we have

$$\langle A, B + C \rangle = \text{tr}(A(B + C)) = \text{tr}(AB) + \text{tr}(AC) = \langle A, B \rangle + \langle A, C \rangle,$$

which proves the second property.

We continue to show the third property. If x is a number, then

$$\langle xA, B \rangle = \text{tr}((xA)B) = x \text{tr}(AB) = x \langle A, B \rangle.$$

On the other hand, we have

$$\langle A, xB \rangle = \text{tr}(A(xB)) = x \text{tr}(AB) = x \langle A, B \rangle,$$

proving the third property.

We finally come to the last property. By the definition of trace, we have

$$\langle A, A \rangle = \text{tr}(AA) = \sum_{i=1}^n \sum_{k=1}^n a_{ik} a_{ki} = \sum_{i=1}^n \sum_{k=1}^n a_{ik}^2 \geq 0,$$

because A is symmetric. Only when $a_{ik} = 0$ for all i, k range in $1 \leq i, k \leq n$, i.e. $A = O$, we have $\langle A, A \rangle = 0$. Hence we conclude the whole proof. $\square$

- (b) Use the dimension formulas. The trace is a linear map, from V to $\mathbf{R}$. Since

$$\dim V = \dim \text{Ker tr} + \dim \text{Im tr},$$

it follows that $\dim W = \dim V - 1$ so $\dim W^\perp = 1$ by Theorem 2.3.

Let I be the unit $n \times n$ matrix. Then $\text{tr}(AI) = \text{tr}(AI)$ for all $A \in V$, so $\text{tr}(AI) = 0$ for $A \in W$. Hence $I \in W^\perp$. Since $W^\perp$ has dimension 1, it follows that I is a basis of $W^\perp$. (Simple?)

10. Let A be a symmetric $n \times n$ matrix. Let $X, Y \in \mathbf{R}^n$ be eigenvectors for A , that is suppose that there exist numbers a, b such that $AX = aX$ and $AY = bY$. Assume that $a \neq b$. Prove that X, Y are perpendicular.

Solution.

Proof. We have $\langle X, AY \rangle = \langle X, bY \rangle = b \langle X, Y \rangle$ and also

$$\langle X, AY \rangle = \langle {}^tAX, Y \rangle = \langle AX, Y \rangle = \langle aX, Y \rangle = a \langle X, Y \rangle,$$

because A is symmetric. So $a \langle X, Y \rangle = b \langle X, Y \rangle$. Since $a \neq b$ it follows that $\langle X, Y \rangle = 0$. $\square$

§3. Bilinear Maps and Matrices

Exercises VI, §3.

1. Let A be $n \times n$ matrix, and assume that A is symmetric, i.e. $A = {}^tA$. Let $\varphi_A : \mathbf{R}^n \times \mathbf{R}^n \rightarrow \mathbf{R}$ be its associated bilinear map. Show that

$$g_A(X, Y) = g_A(Y, X)$$

for all $X, Y \in \mathbf{R}^n$, and thus that g_A is a scalar product, i.e. satisfies conditions **SP 1**, **SP 2**, and **SP 3**.

Solution.

Proof. For all column vectors X, Y we note that tXAY is a 1×1 matrix, so equal to its own transpose. Therefore

$$\varphi_A(X, Y) = {}^tXAY = {}^t({}^tXAY) = {}^tY{}^tA{}^tX = {}^tYAX = \varphi_A(Y, X).$$

□

2.

3.

Chapter VII

Determinants

§1. Determinants of Order 2

§2. 3×3 and $n \times n$ Determinants

Exercises VII, §2.

- 1.
- 2.
- 3.
- 4.
- 5.
- 6.
- 7.
- 8.
- 9.
- 10.
- 11.
- 12.
- 13.
- 14.

§3. The Rank of a Matrix and Subdeterminants

Exercises VII, §3.

- 1.
- 2.
- 3.
- 4.
- 5.
- 6.
- 7.
- 8.
- 9.

§4. Cramer's Rule

Exercises VII, §4.

- 1.

§5. Inverse of a Matrix

Exercises VII, §5.

- 1.
- 2.

§6. Determinants as Area and Volume

Exercises VII, §6.

1. If $g(v, w)$ satisfies the first two axioms of a determinants, prove that

$$g(v, w) = -g(w, v)$$

for all vectors v, w . This fact was used in the uniqueness proof. [*Hint*: Expand $g(v + w, v + w) = 0$.]

solution.

Proof. Let us derive the fact from the second axiom of a determinant. We have

$$\begin{aligned}0 &= g(v + w, v + w) \\&= g(v, v) + g(v, w) + g(w, v) + g(w, w) \\&= g(v, w) + g(w, v)\end{aligned}$$

by linearity in both variables, i.e. the first axiom of a determinant. Thus we see that $g(v, w) = -g(w, v)$. $\square$

2.

3.

4.

Chapter VIII

Eigenvectors and Eigenvalues

§1. Eigenvectors and Eigenvalues

Exercises VIII, §1.

1. Let a be a number $\neq 0$. Prove that the eigenvectors of the matrix

$$\begin{pmatrix} 1 & a \\ 0 & 1 \end{pmatrix}$$

generate a 1-dimensional space, and give a basis for this space.

Solution.

Proof. Let $\begin{pmatrix} x \\ y \end{pmatrix}$ be an eigenvector. Then matrix multiplication shows that we must have $x + ay = \lambda x$ and $y = \lambda y$. If $y \neq 0$ then $\lambda = 1$. Since $a \neq 0$ this contradicts the first equation, so $y = 0$. Then E^1 , which is an eigenvector, forms a basis for the space of eigenvectors. $\square$

- 2.
- 3.
4. Show that if $\theta \in \mathbf{R}$, then the matrix

$$A = \begin{pmatrix} \cos \theta & \sin \theta \\ \sin \theta & -\cos \theta \end{pmatrix}$$

always has an eigenvector in $\mathbf{R}^2$, and in fact that there exists a vector v_1 such that $Av_1 = v_1$. [Hint: Let the first component of v_1 be

$$x = \frac{\sin \theta}{1 - \cos \theta}$$

if $\cos \theta \neq 1$. Then solve for y . What if $\cos \theta = 1$?

solution.

Proof. We are proposed to show that there exists a vector v_1 such that $Av_1 = v_1$. Let $v_1 = \begin{pmatrix} x \\ y \end{pmatrix}$, then we list out the desired expression, that is

$$\begin{cases} \cos \theta \cdot x + \sin \theta \cdot y = x \\ \sin \theta \cdot x - \cos \theta \cdot y = y \end{cases}.$$

We observe that the first equation can be written as

$$(\cos \theta - 1)x + \sin \theta \cdot y = 0$$

Let the first component of v_1 be

$$x = \frac{\sin \theta}{1 - \cos \theta}$$

if $\cos \theta \neq 1$, and we resolve the equations to get $y = 1$. Thus we partly conclude the desired result. (If $\cos \theta = -1$, then $x = 0$ and y ranges in $\mathbf{R}$ except $y = 0$, which also satisfies the desired result.)

If $\cos \theta = 1$, it yields that $y = 0$ and x ranges in $\mathbf{R}$ except $x = 0$, and now we can conclude the desired result. $\square$

5. In Exercise 4, let v_2 be a vector of $\mathbf{R}^2$ perpendicular to the vector v_1 found in that exercise. Show that $Av_2 = -v_2$. Define this to mean that A is a reflection.

solution.

Proof. Let $v_2 = \begin{pmatrix} -y \\ x \end{pmatrix}$ because $\langle v_1, v_2 \rangle = 0$. Then by the listed out equations, we have

$$Av_2 = \begin{pmatrix} \cos \theta & \sin \theta \\ \sin \theta & -\cos \theta \end{pmatrix} \begin{pmatrix} -y \\ x \end{pmatrix} = \begin{pmatrix} \cos \theta \cdot (-y) + \sin \theta \cdot x \\ \sin \theta \cdot (-y) - \cos \theta \cdot x \end{pmatrix} = \begin{pmatrix} y \\ -x \end{pmatrix} = -v_2,$$

which complete the proof. $\square$

6.

7.

§2. The Characteristic Polynomial

Exercises VIII, §2.

- 1.
- 2.
- 3.
- 4.
- 5.
- 6.
- 7.
- 8.
- 9.
- 10.
11. Let A be an invertible matrix. If λ is an eigenvalue of A show that $\lambda \neq 0$ and that λ^{-1} is an eigenvalue of A^{-1} .

Solution.

Proof. That λ is an eigenvalue of A means there exists a vector $v \neq 0$ such that $Av = \lambda v$. Since A is assumed invertible, the kernel of A is O , so $\lambda \neq 0$. Apply A^{-1} to this last equation. We get

$$v = A^{-1}(Av) = A^{-1}(\lambda v) = \lambda A^{-1}v,$$

whence $A^{-1}v = \lambda^{-1}v$, so λ^{-1} is an eigenvalue of A^{-1} . □

- 12.
- 13.
14. Let $A : V \rightarrow V$ be a linear map of V into itself, and let $\{v_1, \dots, v_n\}$ be a basis for V consisting of eigenvectors having distinct eigenvalues $c_1, \dots, c_n$. Show that any eigenvector v of A in V is a scalar multiple of some v_i .

Solution.

Proof. Let v be an eigenvector of A , so that $Av = \lambda v$, $v \neq O$. Since $\{v_1, \dots, v_n\}$ is a basis of V , there exist numbers $a_1, \dots, a_n$ such that

$$v = a_1 v_1 + \dots + a_n v_n.$$

Applying A yields

$$\lambda v = Av = A(a_1 v_1 + \cdots + a_n v_n) = a_1 c_1 v_1 + \cdots + a_n c_n v_n.$$

But we also have

$$\lambda v = \lambda a_1 v_1 + \cdots + \lambda a_n v_n.$$

Subtracting gives

$$a_1(\lambda - c_1)v_1 + \cdots + a_n(\lambda - c_n)v_n = 0.$$

Since $v_1, \dots, v_n$ are linearly independent, we must have

$$a_i(\lambda - c_i) = 0 \quad \text{for all } i = 1, \dots, n.$$

Say $a_1 \neq 0$. Then $\lambda - c_1 = 0$ so $\lambda = c_1$. Since we assumed that the numbers $c_1, \dots, c_n$ are distinct, it follows that $\lambda - c_i \neq 0$ for $j = 2, \dots, n$ whence $a_j = 0$ for $j = 2, \dots, n$. Hence finally $v = a_1 v_1$. This concludes the proof. $\square$

15. Let A, B be square matrices of the same size. Show that the eigenvalues of AB are the same as the eigenvalues of BA .

Solution.

Proof. Let $v \neq 0$ be an eigenvector for AB with eigenvalue λ , so that $ABv = \lambda v$. Then

$$BABv = \lambda Bv.$$

If $Bv \neq 0$ then Bv is an eigenvector for BA with this same eigenvalue λ .

If on the other hand $Bv = 0$, then $\lambda = 0$ because $ABv = \lambda v$ and $v \neq 0$. Furthermore, BA cannot be invertible (otherwise if C is an inverse, $BAC = I$ so B is invertible, which is not the case). Hence there is a vector $w \neq 0$ such that $BAw = 0$, so 0 is also an eigenvalue of BA . This proves that the eigenvalues of AB are also eigenvalues of BA .

There is an even better proof of a more general fact, namely:

The characteristic polynomial of AB is the same as that of BA .

To prove this, suppose first that B is invertible. Since the determinant of a product is the product of the determinants, we get

$$\det(xI - AB) = \det(B(xI - AB)B^{-1}) = \det(xI - BABB^{-1}) = \det(xI - BA).$$

This proves the theorem assuming B invertible. But the identity to be proved

$$\det(xI - AB) = \det(xI - BA)$$

is a polynomial identity, in which we can consider the components of A and B as “variables”, so if the identity is true when A is fixed and $B = (b_{ij})$ is “variable”, i.e. the polynomial identity holds for infinitely many distinct matrices B , then it is true for all B . A matrix with “variable” components is invertible, so the previous argument applies to conclude the proof. $\square$

§3. Eigenvalues and Eigenvectors of Symmetric Matrices

Exercises VIII, §3.

- 1.
- 2.
- 3.

§4. Diagonalization of a Symmetric Linear Map

Exercises VIII, §4.

- 1.
- 2.
3. Let V be a finite dimensional vector space with a positive definite scalar product. Let $A : V \rightarrow V$ be a symmetric linear map. We say that A is **positive definite** if $\langle Av, v \rangle > 0$ for all $v \in V$ and $v \neq O$. Prove:
 - (a) If A is positive definite, then all eigenvalues are > 0 .
 - (b) If A is positive definite, then there exists a symmetric linear map B such that $B^2 = A$ and $BA = AB$. What are the eigenvalues of B ? [*Hint*: Use a basis of V consisting of eigenvectors.]

Solution.

- (a) *Proof.* Let v be an eigenvector $\neq O$ with eigenvalue λ . Then $\langle Av, v \rangle = \langle \lambda v, v \rangle = \lambda \langle v, v \rangle$. Since $\langle v, v \rangle > 0$ and $\langle Av, v \rangle > 0$, it follows that $\lambda > 0$. □
- (b) *Proof.* Pick a basis of V consisting of eigenvectors. The vector space V can then be identified as the space of coordinate vectors with respect to this basis. The matrix of A then is a diagonal matrix, whose diagonal elements are the eigenvalues, and are therefore positive. We can then let B have diagonal elements equal to the square roots of those of A . □
4. We say that A is **positive semidefinite** if $\langle Av, v \rangle \geq 0$ for all $v \in V$. Prove the analogues of (a), (b) of Exercise 3 when A is only assumed semidefinite. Thus the eigenvalues are ≥ 0 , and there exists a symmetric linear map B such that $B^2 = A$.

solution.

- (a) *Proof.* Let v be an eigenvector $\neq O$ with eigenvalue λ . Then $\langle Av, v \rangle = \langle \lambda v, v \rangle = \lambda \langle v, v \rangle$. Since $\langle v, v \rangle \geq 0$ and $\langle Av, v \rangle \geq 0$, it follows that $\lambda \geq 0$. □

(b) *Proof.* Pick a basis of V consisting of eigenvectors. The vector space V can then be identified as the space of coordinate vectors with respect to this basis. The matrix of A then is a diagonal matrix, whose diagonal elements are the eigenvalues, and are therefore nonnegative. We can then let B have diagonal elements equal to the square roots of those of A . $\square$

5. Assume that A is symmetric positive definite. Show that A^2 and A^{-1} are symmetric positive definite.

Solution.

Proof. From ${}^t(AA) = {}^tA{}^tA = AA$, it follows that A^2 is symmetric. Furthermore, for $v \neq 0$,

$$\langle A^2v, v \rangle = \langle Av, {}^tAv \rangle = \langle Av, Av \rangle > 0,$$

because $Av \neq 0$ since $\langle Av, v \rangle > 0$.

Since ${}^tA^{-1} = A^{-1}$, it follows that A^{-1} is symmetric. Since A is invertible, a given v can be written $v = Aw$ for some w (namely $w = A^{-1}v$). Then

$$\langle A^{-1}v, v \rangle = \langle A^{-1}Aw, Aw \rangle = \langle w, Aw \rangle = \langle Aw, w \rangle > 0.$$

Hence A^{-1} is positive definite. $\square$

6. Let $U : \mathbf{R}^n \rightarrow \mathbf{R}^n$ be a linear map, and let $\langle \cdot, \cdot \rangle$ denote the usual scalar (dot) product. Show that the following conditions are equivalent:

- (i) $\|Uv\| = \|v\|$ for all $v \in \mathbf{R}^n$.
- (ii) $\langle Uv, Uw \rangle = \langle v, w \rangle$ for all $v, w \in \mathbf{R}^n$.
- (iii) U is invertible, and ${}^tU = U^{-1}$.

[Hint: For (ii), use the identity

$$\langle (v+w), (v+w) \rangle - \langle (v-w), (v-w) \rangle = 4\langle v, w \rangle,$$

and similarly with a U in front of each vector.] When U satisfies any one (and hence all) of these conditions, then U is called **unitary**. The first condition says that U preserves the norm, and the second says that U preserves the scalar product.

Solution.

Proof. Assume (i). From the identity in the hint, we get

$$\begin{aligned} 4\langle Uv, Uw \rangle &= \langle U(v+w), U(v+w) \rangle - \langle U(v-w), U(v-w) \rangle \\ &= \|U(v+w)\|^2 - \|U(v-w)\|^2 \\ &= \|v+w\|^2 - \|v-w\|^2 \\ &= \langle (v+w), (v+w) \rangle - \langle (v-w), (v-w) \rangle \\ &= 4\langle v, w \rangle. \end{aligned}$$

Hence $\langle Uv, Uw \rangle = \langle v, w \rangle$. The converse is immediate.

Assume (i). Then $\text{Ker } U = O$ because if $Uv = O$ then $\|v\| = 0$ so $v = O$. But a linear map with O kernel from a finite dimensional vector space into itself is invertible, so U is invertible. Also, for all $v, w \in V$,

$$\langle Uv, Uw \rangle = \langle v, {}^tUUw \rangle \quad \text{and also equals} \quad \langle v, w \rangle \quad \text{by hypothesis.}$$

Hence ${}^tUU = I$ so ${}^tU = U^{-1}$. Conversely, from ${}^tU = U^{-1}$ we get

$$\|Uv\| = \langle Uv, Uv \rangle = \langle {}^tUUv, v \rangle = \langle v, v \rangle = \|v\|,$$

so U satisfies (i). □

7. Let $A : \mathbf{R}^n \rightarrow \mathbf{R}^n$ be an invertible linear map.

- (i) Show that tAA is symmetric positive definite.
- (ii) By Exercise 3b, there is a symmetric positive definite B such that $B^2 = {}^tAA$. Let $U = AB^{-1}$. Show that U is unitary.
- (iii) Show that $A = UB$.

Solution.

- (i) *Proof.* Since ${}^t({}^tAA) = {}^tA{}^tA = {}^tAA$, it follows that tAA is symmetric. Furthermore, for $v \neq O$, we have

$$\langle {}^tAAv, v \rangle = \langle Av, Av \rangle > 0$$

because A is invertible, $Av \neq O$. Hence tAA is positive definite. □

- (ii) *Proof.* Let $U = AB^{-1}$ where $B^2 = {}^tAA$. Then

$$\begin{aligned} \langle Uv, Uv \rangle &= \langle AB^{-1}v, AB^{-1}v \rangle \\ &= \langle B^{-1}v, {}^tAAB^{-1}v \rangle \\ &= \langle B^{-1}v, Bv \rangle \\ &= \langle {}^tBB^{-1}v, v \rangle \\ &= \langle v, v \rangle, \end{aligned}$$

because B is symmetric. Hence U is unitary. □

- (iii) *Proof.* Since $U = AB^{-1}$, we have

$$UB = AB^{-1}B$$

and hence $A = UB$. □

8. Let B be symmetric positive definite and also unitary. Show that $B = I$.

Solution.

Proof. Let v be a non-zero eigenvector $\lambda > 0$. Then

$$\langle Bv, Bv \rangle = \lambda^2 \langle v, v \rangle = \langle v, v \rangle$$

because B is unitary. Hence $\lambda^2 = 1$. Hence $\lambda = \pm 1$, and since λ is positive, $\lambda = 1$. Since V has a basis consisting of eigenvectors, it follows that $B = I$. □